Zhiyu Xu

Ph.D. Student in Statistics, Columbia University

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EDUCATION

Columbia University **Sep 2024 – Present**
Ph.D. in Statistics; GPA: 4.10/4.00 Expected Jun 2029

Tsinghua University **Sep 2020 – Jun 2024**
B.S. Mathematics and Physics; B.E. Energy and Power Engineering GPA: 3.97/4.00

RESEARCH INTERESTS

Latent-variable models, probabilistic machine learning, and psychometrics, with applications to educational assessment and large language model evaluation.

RESEARCH

Latency-Response Theory: Evaluating LLMs with Accuracy and Chain-of-Thought Length 2026
Zhiyu Xu, Jia Liu, Yixin Wang, Yuqi Gu | Paper | Python code
Annals of Applied Statistics, Special Issue in Statistics and Artificial Intelligence

- Developed a joint latent-variable model for response accuracy and chain-of-thought length, with estimation via stochastic-approximation EM and data-driven initialization.
- Established identifiability and asymptotic posterior normality; characterized the role of ability–speed correlation in estimation precision.
- Examined prediction, calibration, and adaptive testing across four mathematical reasoning benchmarks; implemented the method and released code and processed data.

CRYSTAL: Cluster-geometry-inspired Spectral Algorithm for Binary Latent Feature Models 2026
Chengzhu Huang, Yuqi Gu, **Zhiyu Xu** (alphabetical order)
Major revision, *Journal of the American Statistical Association*, Theory and Methods

- Developed a spectral algorithm for binary latent feature estimation in linear and nonlinear models, allowing overlapping group membership; established identifiability for both model classes.
- Studied latent feature recovery in simulations and applications to educational assessment and species abundance data.

Scalable Text-Embedding-Informed Cognitive Diagnosis of Large Language Models 2026
Jia Liu, **Zhiyu Xu**, Yuqi Gu | Preprint | Revise and resubmit, *Psychometrika*

- Developed a cognitive diagnosis framework that jointly estimates latent skill mastery and the skills required by evaluation items.
- Incorporated text-embedding-informed priors into stochastic-approximation inference; studied parameter recovery in simulations and skill profiles on MATH Level 5.

Covariate-Assisted Grade of Membership Models via Shared Latent Geometry 2026
Zhiyu Xu, Yuqi Gu | Preprint | R code

- Developed a likelihood-free spectral estimator that combines binary responses and covariates through shared simplex geometry and heteroskedastic PCA.
- Established identifiability conditions and finite-sample entrywise error bounds; studied how covariate information contributes to mixed membership recovery.
- Implemented the estimator in R and studied estimation and prediction through simulations and an application to student proficiency profiles in TIMSS data.

HONORS

ASA Statistical Learning and Data Science Student Paper Award 2026
NSF SLDS 2026 Travel Support Award 2026

TALKS

SLDS 2026: Inference and Intelligence, Brooklyn (upcoming) Nov 2026
Joint Statistical Meetings (JSM), Boston Aug 2026
International Meeting of the Psychometric Society (IMPS), Seoul Jul 2026
Joint Statistical Meetings (JSM), Nashville Aug 2025

TEACHING EXPERIENCE

Graduate Teaching Assistant, Columbia University 2024 – 2026
Statistical Machine Learning; Probability Theory; Linear Regression; Applied Statistics (Ph.D. course)

COMPUTATIONAL SKILLS

Languages: Python, R, C++, MATLAB
Libraries and tools: NumPy, SciPy, pandas, PyTorch, JAX, Git, Slurm, LaTeX